MTH319 2022S2 Exam

Feel free to contact me at maths_and_programming@outlook.sg or @geodome on Telegram if you need tuition.

I provide undergraduate level tuition for the following SUSS units:

- ENG301 Microprocessor Progtramming
- ICT133 Structured Programming
- ICT162 Object Oriented Programming
- ICT235 Data Structures and Algorithms
- ICT323 Introduction to Parallel Computing
- ICT325 Design and Analysis of Algorithms
- ICT330 Database Management Systems
- MTD215 Application of C++ in Multimedia
- MTH105 Fundamentals of Mathematics
- MTH109 Calculus
- MTH207 Linear Algebra
- MTH210 Fundamentals of Probability
- MTH212 Statistical Analysis
- MTH240 Engineering Mathematics 1
- MTH316 Multivariable Calculus
- MTH321 Engineering Mathematics 2
- MTH355 Basic Mathematical Optimisation

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MTH319

Timed Online Assignment – July Semester 2021

Mathematical Methods I

Friday, 19 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and must not exceed 500MB in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

In an online game, there are three items that you can equip to increase your defence points against magical and physical attacks. The number of each item you can equip are x, y and z respectively. The contributions and price of each item are shown below:

ITEM	Magical Defence Points	Physical Defence Points	Price per Item (Coins)
Leather Scale (x)	1	4	40
Enchanted Ruby (y)	3	-2	20
Phoenix Feather (z)	5	3	100

Table Q1

(a) In order to clear a boss fight, you need to increase your magical defence by 20 points and physical defence by 10 points. If the total number of coins you intend to spend is 300, show that the system of linear equations that describes the problem is:

$$A\mathbf{v} = \mathbf{b} \longrightarrow \begin{bmatrix} 1 & 3 & 5 \\ 4 & -2 & 3 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 15 \end{bmatrix}.$$

(4 marks)

(b) Using row operations on the augmented matrix [A|**b**], solve the matrix equation in Question 1(a) for the number of units of each item. Describe the system in terms of its linear independence, consistency and rank.

(12 marks)

Question 2

An online seller of home-baked cakes is trying to apply some data analytics in order to improve her sales. Table Q2 below shows the price-volume relationship of her daily sales.

Price, P ($)	Sold Volume per day, V (cakes)
9	4
8	5
6	8
4	10

Table Q2

(a) Using linear regression by minimizing the sum of squared errors, show that the predictive model for the volume function of price is:

$$V(P) = -1.24P + 15.1.$$

Hence, if the seller prices each cake at $4.90, how many cakes can she sell approximately in one day?

(13 marks)

(b) If a new data point given by (P, V) = ($2.5, 12) is added, how would the linear fit change? Explain briefly.

(3 marks)

(c) The linear equation given in Question 2(a) can also be expressed by the vector equation:

$$\vec{r} = \vec{u} + t\vec{v}.$$

If the position vector **u** is given below, determine the vector **v** such that the vector equation is fully defined.

$$\vec{u} = \begin{bmatrix} 0 \\ 15.1 \end{bmatrix}$$

(5 marks)

Question 3

Given the matrix A below:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Compute the eigendecomposition of A. For each eigenvalue, describe its eigenspace.

(16 marks)

Question 4

Figure Q4 below shows the use of a copper heat sink (of mass, m, specific heat capacity, c and exposed surface area, A) to transfer heat away from a CPU chip in order to prevent overheating of the chip. The CPU chip generates heat at a constant Q Watts during operation which has to be dissipated to the surroundings at temperature T_s.

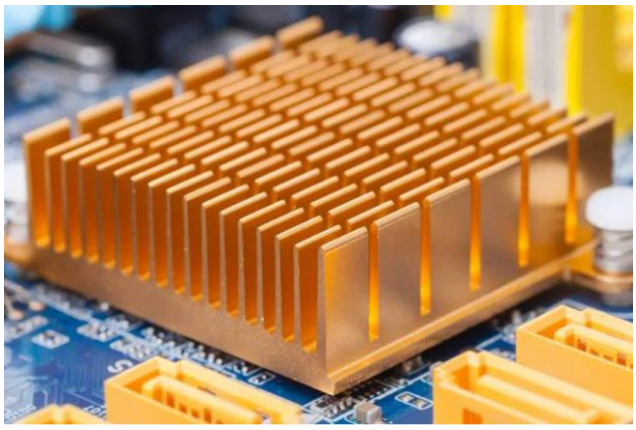

Figure Q4: Heat sink for CPU heat dissipation

Source: https://www.arrow.com/en/research-and-events/articles/choosing-the-right-heat-sink

(a) The CPU is off for a long time and turned on at t = 0. By applying the conservation of energy to the heat sink, derive the mathematical model that governs the heat transfer phenomenon and show that it results in the initial value problem:

$$T' + \frac{UA}{mc}T = \frac{1}{mc}(Q + UAT_s), \quad T(0) = T_s$$

where T(t) represents the temperature function of the heat sink and U is the overall heat transfer coefficient. State your assumptions clearly.

(10 marks)

(b) Determine the solution of the initial value problem and show that the temperature function of the heat sink is:

$$T(t) = -\frac{Q}{UA}e^{-\frac{UA}{mc}t} + \frac{Q}{UA} + T_s.$$

(11 marks)

(c) State the transient and the steady-state expressions of temperature function. Are the expressions logical? Explain.

(6 marks)

(d) Given the following parameters, determine the steady-state temperature of the heat sink during operation of the CPU. Discuss on the suitability of this heat sink for cooling the CPU.

$U = 13\ W/m^2.K$, $A = 0.02\ m^2$, $m = 0.8\ kg$,
$c = 390\ J/kg.K$, $Q = 130\ W$, $T_s = 24\ °C$.

(5 marks)

Question 5

Solve the system of ODEs given below. What happens to the solutions as time t approaches infinity?

$$\begin{aligned} x' &= -2x - 4y \\ y' &= x - 6y \end{aligned}, \quad x(t=0) = 7, \ y(t=0) = 3$$

(15 marks)

----- END OF PAPER -----

1(a) Equation for balancing Mental Defence Points:

$x + 3y + 5z = 20$

Equation for balancing Physical Defence Points:

$4x - 2y + 3z = 10$

Equation for expenditure:

$40x + 20y + 100z = 300$

$\Rightarrow 2x + y + 5z = 15$

$$\therefore \begin{bmatrix} 1 & 3 & 5 \\ 4 & -2 & 3 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 15 \end{bmatrix}$$

(b)

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & 20 \\ 4 & -2 & 3 & 10 \\ 2 & 1 & 5 & 15 \end{array}\right] \xrightarrow[r_3 = r_3 - 2r_1]{r_2 = r_2 - 4r_1} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 20 \\ 0 & -14 & -17 & -70 \\ 0 & -5 & -5 & -25 \end{array}\right] \xrightarrow[r_3 = r_3 - \frac{5}{14} r_2]{r_2 = -\frac{1}{14} r_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 20 \\ 0 & 1 & 17/14 & 5 \\ 0 & 0 & 15/14 & 0 \end{array}\right]$$

$$\downarrow r_3 = \frac{14}{15} r_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{array}\right] \xleftarrow{r_1 = r_1 - 3r_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 20 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{array}\right] \xleftarrow[r_1 = r_1 - 5r_3]{r_2 = r_2 - \frac{17}{14} r_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 20 \\ 0 & 1 & 17/14 & 5 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}$$

The system is consistent since there exists one solution

There are 3 pivots, so the rank is 3.

The system of equation is linearly independent

2(a) $\begin{bmatrix} m \\ c \end{bmatrix} = (A^TA)^{-1}A^TV$ where $V = \begin{bmatrix} 4 \\ 5 \\ 8 \\ 10 \end{bmatrix}$, $A = \begin{bmatrix} 9 & 1 \\ 8 & 1 \\ 6 & 1 \\ 4 & 1 \end{bmatrix}$

$A^TA = \begin{bmatrix} 197 & 27 \\ 27 & 4 \end{bmatrix}$ $\quad (A^TA)^{-1} = \frac{1}{59}\begin{bmatrix} 4 & -27 \\ -27 & 197 \end{bmatrix}$

$A^TV = \begin{bmatrix} 164 \\ 27 \end{bmatrix}$

$$\begin{bmatrix} m \\ c \end{bmatrix} = (A^TA)^{-1}(A^TV)$$
$$= \frac{1}{59}\begin{bmatrix} 4 & -27 \\ -27 & 197 \end{bmatrix}\begin{bmatrix} 164 \\ 27 \end{bmatrix}$$
$$= \frac{1}{59}\begin{bmatrix} -73 \\ 891 \end{bmatrix}$$

$$\therefore v(p) = -\frac{73}{59}p + \frac{891}{59}$$
$$= -1.24p + 15.1$$

$v(4.90) = 9.038983$

She can sell 9 cupcakes a day at \$4.90

(b) $v(2.5) = 12$

This means the new data point actually lies on the linear fit, so adding the new data point doesn't change the regression line.

(c) $$\underset{\sim}{x} = \begin{bmatrix} p \\ v \end{bmatrix}$$
$$= \begin{bmatrix} p \\ -1.24p + 15.1 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ 15.1 \end{bmatrix} + p\begin{bmatrix} 1 \\ -1.24 \end{bmatrix}$$

3. let $\det(A-\lambda I)=0$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)[(2-\lambda)^2-1] = 0$$

$$(1-\lambda)(\lambda^2-4\lambda+3) = 0$$

$$-(\lambda-1)(\lambda-1)(\lambda-3) = 0$$

$$\therefore \lambda_{1,2} = 1$$

$$\therefore \lambda_{2,1} = 3$$

case $\lambda_{1,2} = 1$:

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_3 \\ x_3 \end{bmatrix} = x_1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_3\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{eigenspace} = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}\right\} \quad v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

case $\lambda_{2,1} = 3$:

$$\left[\begin{array}{ccc|c} -2 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ x_3 \\ x_3 \end{bmatrix} = x_3\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\therefore \text{eigenspace} = \text{span}\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right\} \quad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

4(a) Assuming the CPU chip is hotter than surroundings

$$Q - Q_{out} = \frac{dE}{dt} = mc\frac{dT}{dt} = mcT'$$

$Q =$ constant heat generation

$Q_{out} = UA(T - T_s)$.

$$Q - UA(T - T_s) = mcT'$$

$$Q + UAT_s = mcT' + UAT$$

$$\therefore \frac{1}{mc}(Q + UAT_s) = T' + \frac{UA}{mc}T$$

Since the CPU was switched off for a long time prior to switching on,

$T(0) = T_s$

(b) $$T' + \frac{UA}{mc}T = \frac{Q}{mc} + \frac{UA}{mc}T_s$$

use integrating factor $\mu(t) = e^{\int \frac{UA}{mc}dt} = e^{\frac{UA}{mc}t}$

$$T'e^{\frac{UA}{mc}t} + \frac{UA}{mc}e^{\frac{UA}{mc}t} = \frac{Q}{mc}e^{\frac{UA}{mc}t} + \frac{UA}{mc}e^{\frac{UA}{mc}t}T_s$$

$$\frac{d}{dt}\left(Te^{\frac{UA}{mc}t}\right) = \frac{Q}{mc}e^{\frac{UA}{mc}t} + \frac{UA}{mc}e^{\frac{UA}{mc}t}T_s$$

$$Te^{\frac{UA}{mc}t} = \int \frac{Q}{mc}e^{\frac{UA}{mc}t} + \frac{UA}{mc}e^{\frac{UA}{mc}t}T_s \, dt$$

$$= \frac{Q}{UA}e^{\frac{UA}{mc}t} + e^{\frac{UA}{mc}t}T_s + C$$

$$T = \frac{Q}{UA} + T_s + \frac{C}{e^{\frac{UA}{mc}t}}$$

$$T_s = T(0) = \frac{Q}{UA} + T_s + C$$

$$C = -\frac{Q}{UA}$$

$$T(t) = \frac{Q}{UA} + T_s - \frac{Q}{UA}\cdot\frac{1}{e^{\frac{UA}{mc}t}}$$

$$= T_s + \frac{Q}{UA}\left[1 - e^{-\frac{UA}{mc}t}\right]$$

4(c) Transient component $T_{transient} = -\frac{Q}{UA} e^{-\frac{UA}{mc}t}$

Steady-state component $T_{steady} = \frac{Q}{UA} + T_s$

As t approaches ∞, the transient term reduces to 0, so the steady state component sets the temperature. The solutions are logical.

(d) $T_{steady} = \frac{Q}{UA} + T_s$

$= \frac{130}{13(0.02)} + 24$

$= 524°C$

5. $x' = -2x - 4y$
$y' = x - 6y$

$\Leftrightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -2 & -4 \\ 1 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

let $A = \begin{bmatrix} -2 & -4 \\ 1 & -6 \end{bmatrix}$

let $\det(A - \lambda I) = 0$

$$\begin{vmatrix} -2-\lambda & -4 \\ 1 & -6-\lambda \end{vmatrix} = 0$$

$(-2-\lambda)(-6-\lambda) - (-4)(1) = 0$

$(\lambda+2)(\lambda+6) + 4 = 0$

$\lambda^2 + 8\lambda + 16 = 0$

$(\lambda+4)^2 = 0$

$\lambda = -4$

$\therefore \lambda_{1,2} = -4$

$$\left[\begin{array}{cc|c} -2-\lambda & -4 & 0 \\ 1 & -6-\lambda & 0 \end{array}\right] \xrightarrow{\lambda=-4} \left[\begin{array}{cc|c} 2 & -4 & 0 \\ 1 & -2 & 0 \end{array}\right] \xrightarrow[r_2 = r_1 - \frac{1}{2}r_1]{r_1 = \frac{1}{2}r_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad \therefore v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 2 & -4 & 2 \\ 1 & -2 & 1 \end{array}\right] \xrightarrow{\text{swap } r_1 \text{ and } r_2} \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 2 & -4 & 2 \end{array}\right] \xrightarrow{r_2 = r_2 - 2r_1} \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 + 1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \therefore v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}, \quad J = \begin{bmatrix} -4 & 1 \\ 0 & -4 \end{bmatrix}$$

$\therefore$ Jordan Normal Form of $A = PJP^{-1}$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -4 & 1 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = Pe^{Jt}P^{-1}\vec{x}_0$$

$$= \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} e^{-4t} & te^{-4t} \\ 0 & e^{-4t} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2te^{-4t} + 7e^{-4t} \\ te^{-4t} + 3e^{-4t} \end{bmatrix} = e^{-4t} \begin{bmatrix} 2t+7 \\ t+3 \end{bmatrix}$$

5. $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = e^{-4t} \begin{bmatrix} 2t+7 \\ t+3 \end{bmatrix}$

$$\lim_{t\to\infty} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \lim_{t\to\infty} e^{-4t} \begin{bmatrix} 2t+7 \\ t+3 \end{bmatrix}$$

$$= \lim_{t\to\infty} \begin{bmatrix} (2t+7)/e^{4t} \\ (t+3)/e^{4t} \end{bmatrix}$$

$\frac{\infty}{\infty}$ form

L'Hopital Rule

$$= \lim_{t\to\infty} \begin{bmatrix} \frac{2}{4e^{4t}} \\ \frac{1}{4e^{4t}} \end{bmatrix}$$

$$= \lim_{t\to\infty} \frac{1}{e^{4t}} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$